Building intelligent Global Health and
Humanitarian Response Technologies

Week 8:
Advice at the Wrong Time

Context Why Doctors *Ignore* Good Advice
Engineering Integration Without *Interruption*

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EPFL ARIADNE LABS ASHOKA UNIVERSITY WORLD ECONOMIC FORUM Rwanda Centre for the Fourth Industrial Revolution

When does the cure become *worse* than the disease?

- A test can begin a chain of harm long before any disease is found.
- An alert can be clinically correct and still make care worse.
- “Doing nothing” is sometimes neglect—but sometimes it is expertise.
- AI can reduce low-value care, or it can manufacture more recommendations, more anxiety, and more work.

EPFL

Why Doctors Ignore (Good) Advice

Lecturer: *Dr. Veronique Suttels MD, PhD*

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CONTEXT

Outline

01 **Threshold thinking and bias**

02 **Why advice is ignored**

- Too much medicine (Low-value care)
- How alerts become background noise
- Other reasons

03 **Co-Designing AI for medical advice**

Threshold thinking & bias

CONTEXT

SYSTEM 1
Intuiition & instinct

95%

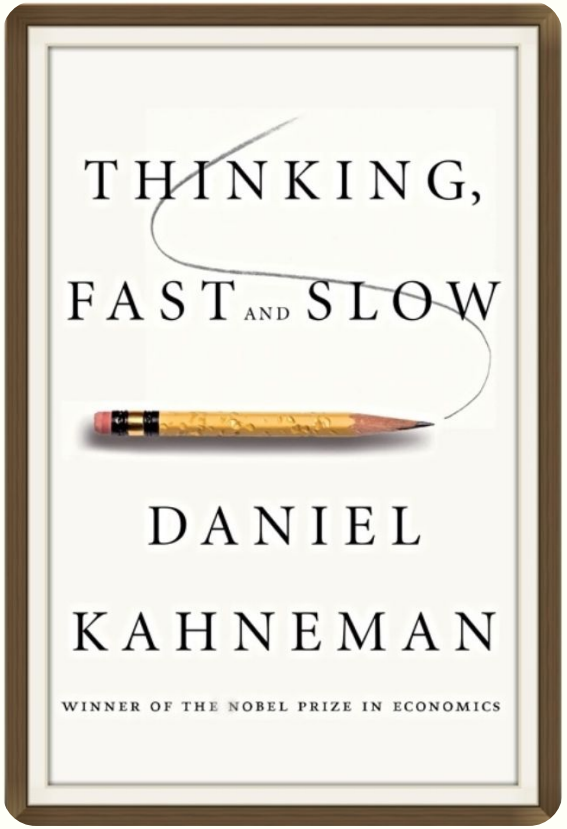
Unconscious
Fast
Associative
Automatic pilot



SYSTEM 2
Rational thinking

5%

Takes effort
Slow
Logical
Lazy
Indecisive

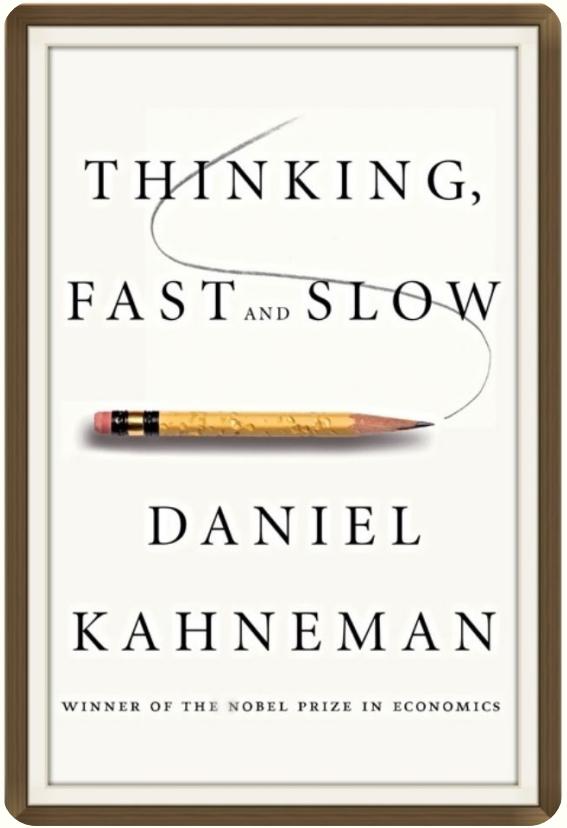


Kahneman, 2011

Threshold thinking & bias

CONTEXT

A bat and a ball cost $1.10 in total.
The bat costs $1.00 more than the ball.
How much does the ball cost?

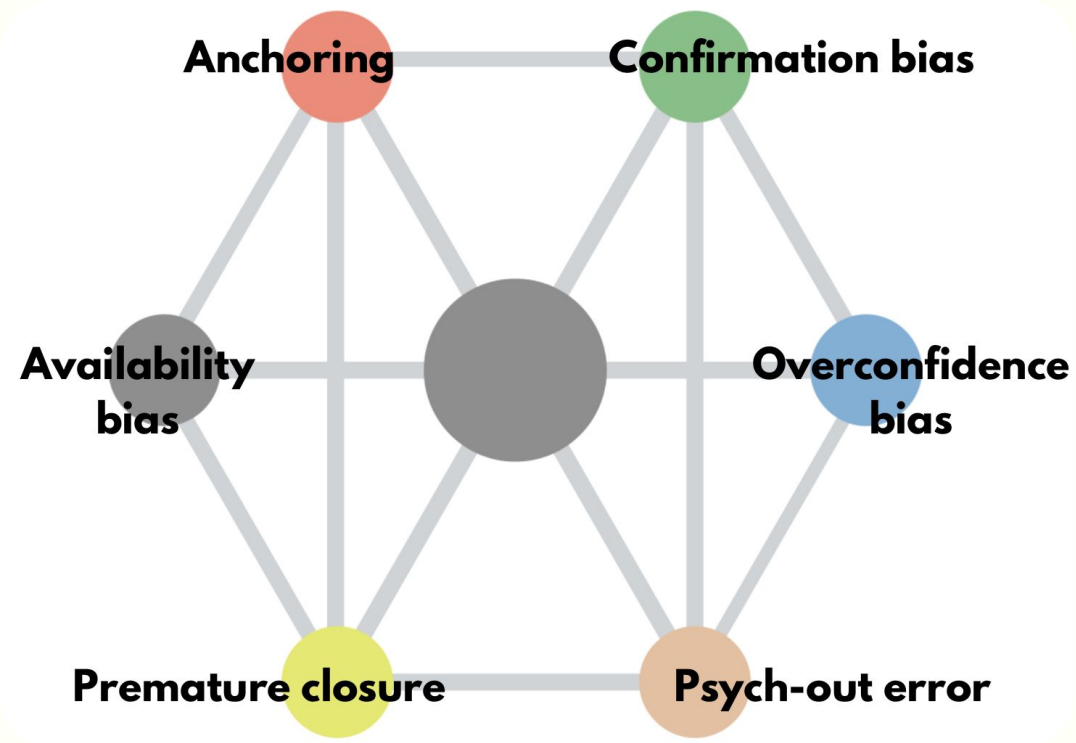


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Threshold thinking & bias

CONTEXT

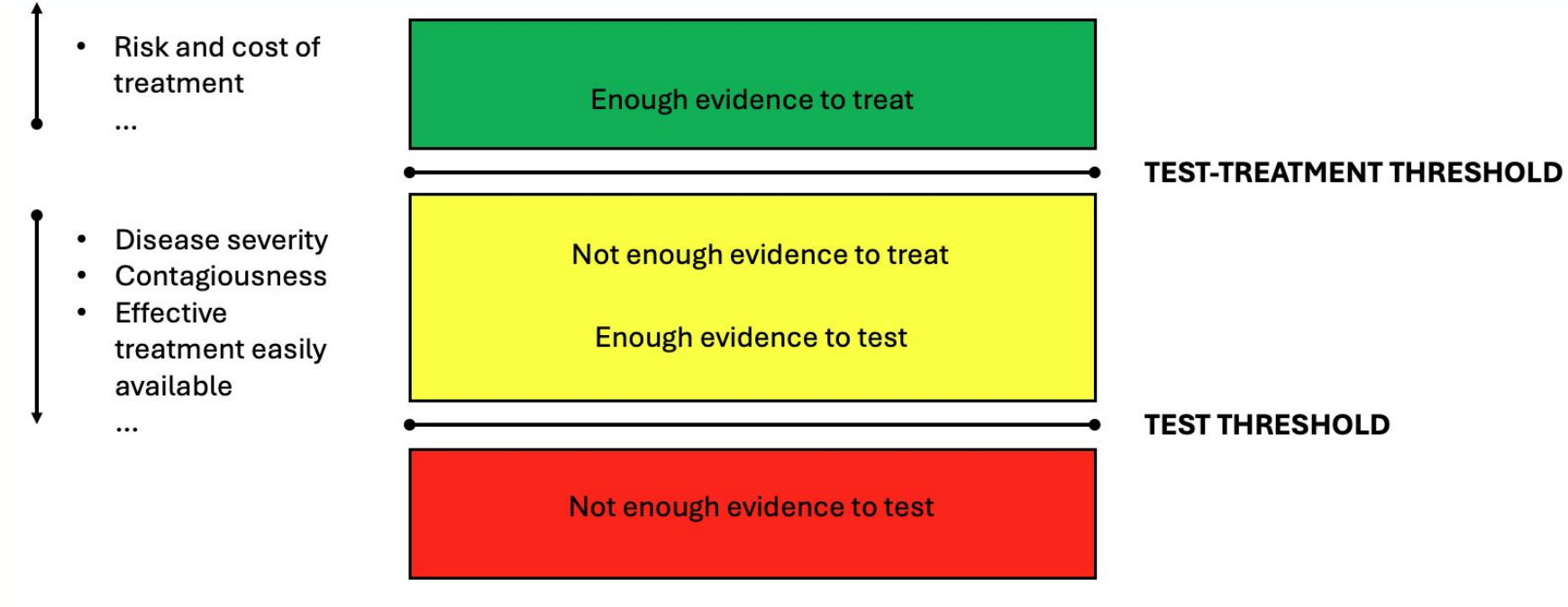
Any random clinician is a web of bias…



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Threshold thinking & bias

CONTEXT



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Pauker and Kassirer, NEJM, 1980

Threshold thinking & bias

- Humans are error prone
- We need advice
- Why do we ignore it?

Why advice is ignored

CONTEXT

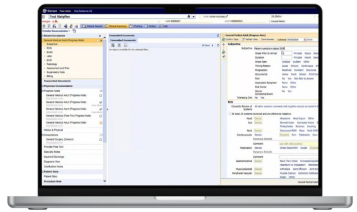
Whas it good advice?

- Too much medicine
- Low-value care

How was this advice dispensed?

- Interruptive alert
- Non-interruptive alert

Other reasons



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Was it good advice?

CONTEXT

Too much medicine



Imagine

You are a 35 year old patient and you need an ultrasound to screen for liver cancer (because you are at high risk). The hospital offers to do a thyroid ultrasound for a small extra fee to screen for thyroid cancer.

Do you accept? Why (not)?

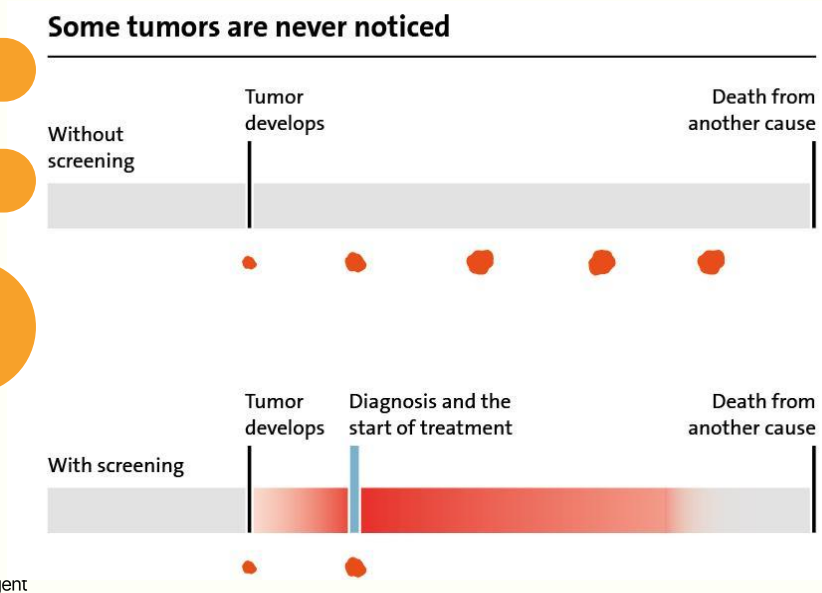
Was it good advice?

CONTEXT

Too much medicine

Effects of screening

- Overdiagnosis
- Overtreatment
- Life not extended



Terminology

incidentaloma

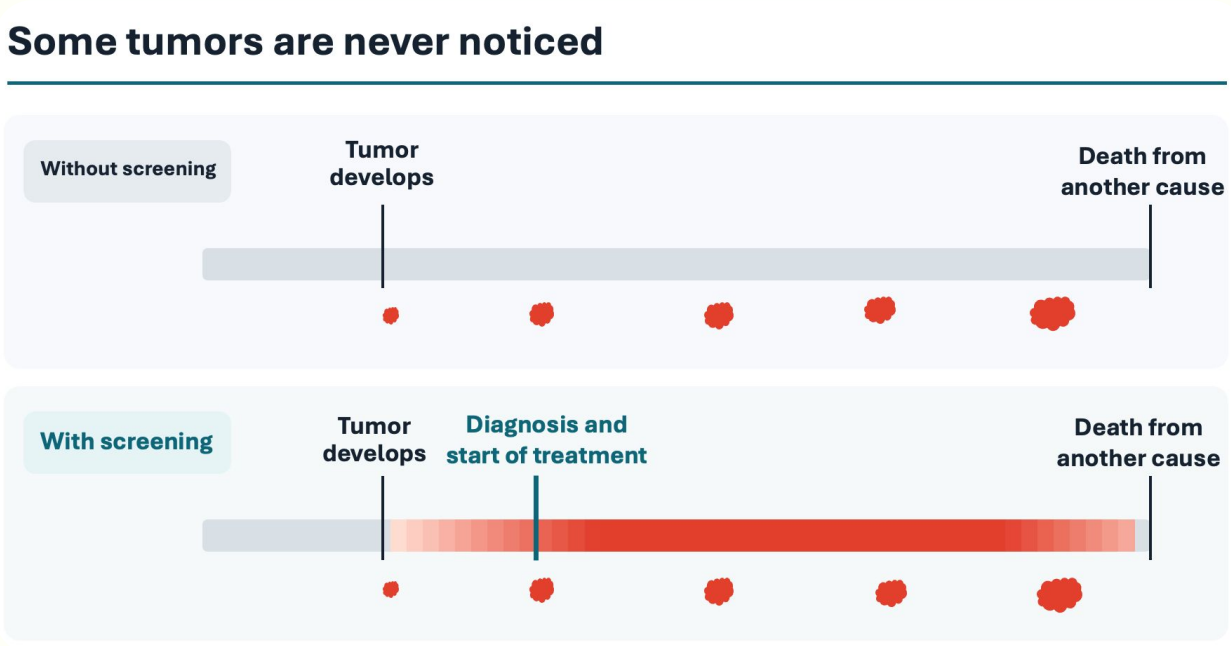
- Real disease, correctly detected
- Would never cause symptoms or death
- Often detected when looking for something else

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Was it good advice?

CONTEXT

Too much medicine



Terminology

incidentaloma

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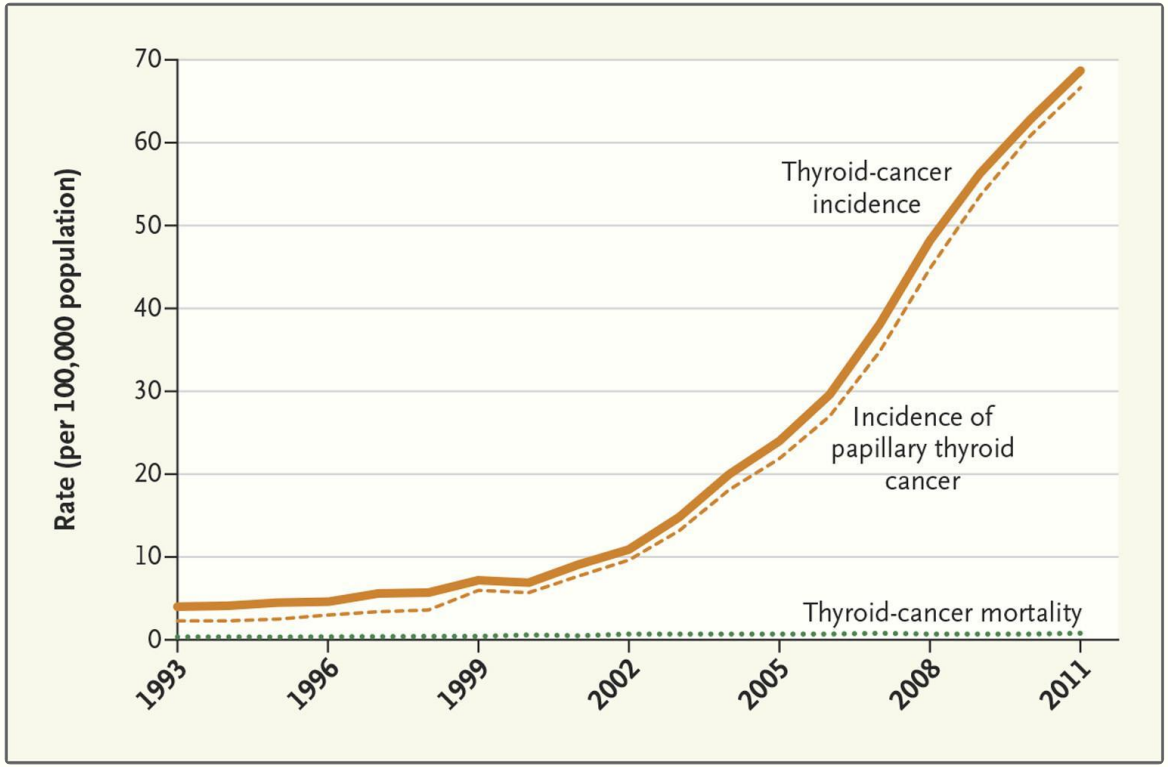
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Was it good advice?

CONTEXT

Too much medicine





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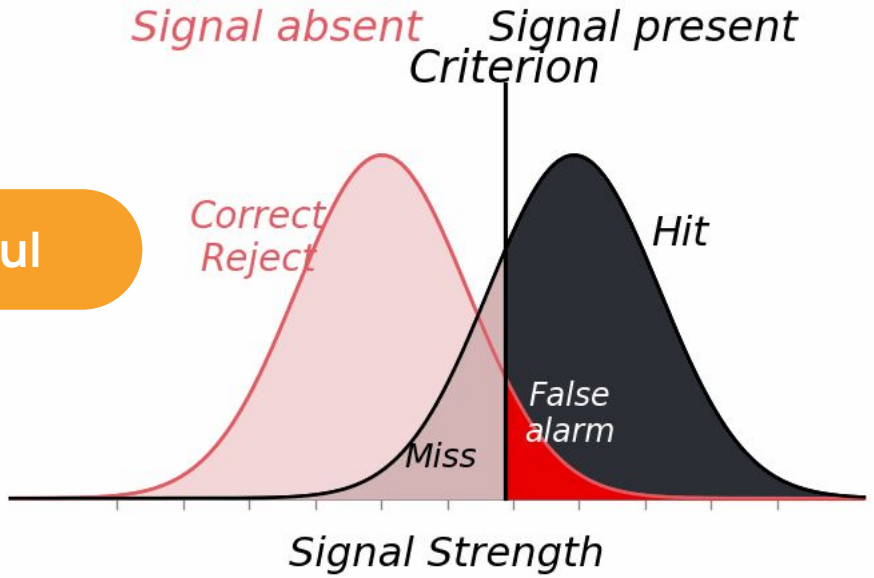
Ahn, NEJM 2014

Was it good advice?

CONTEXT

Too much medicine

more sensitive ≠ more useful



Too much medicine

«Harmless» findings are not truly harmless

Low-value care and the cascade problem

CONTEXT

What best defines low-value care?

A. Care provided using outdated medical technology augmenting costs

B. Care offering little or no clinical benefit or even harm, relative to its costs

C. Care delivered by inexperienced clinicians resulting in more costs

D. Care that produces an incorrect diagnosis augmenting costs

Low-value care and the cascade problem

CONTEXT

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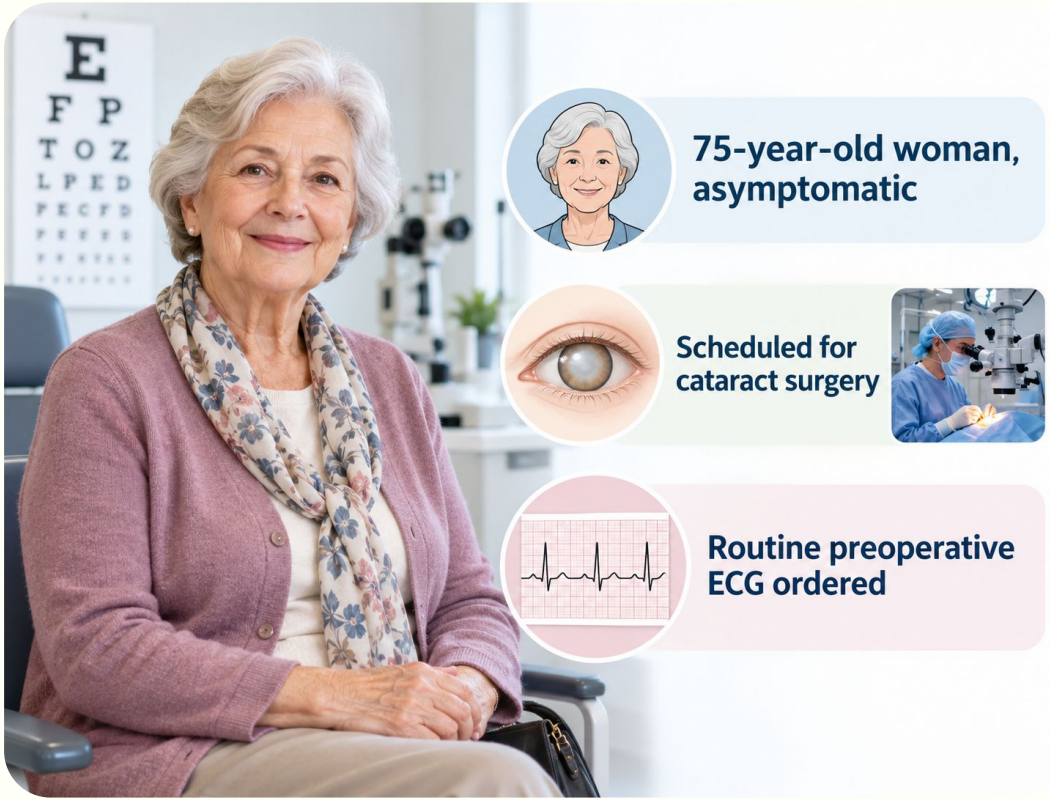
D. Care that produces an incorrect diagnosis augmenting costs

Was it good advice?

CONTEXT

Low value care and its cascade

Clinical case study



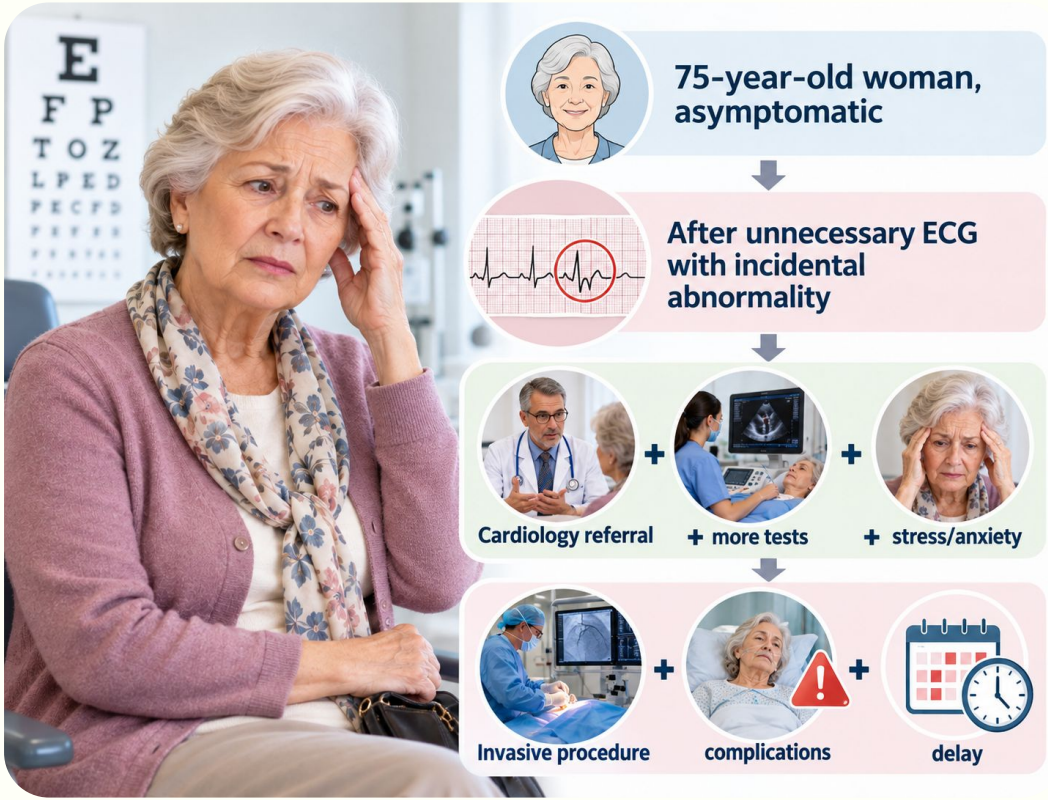
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Was it good advice?

CONTEXT

Low value care and its cascade

Clinical case study

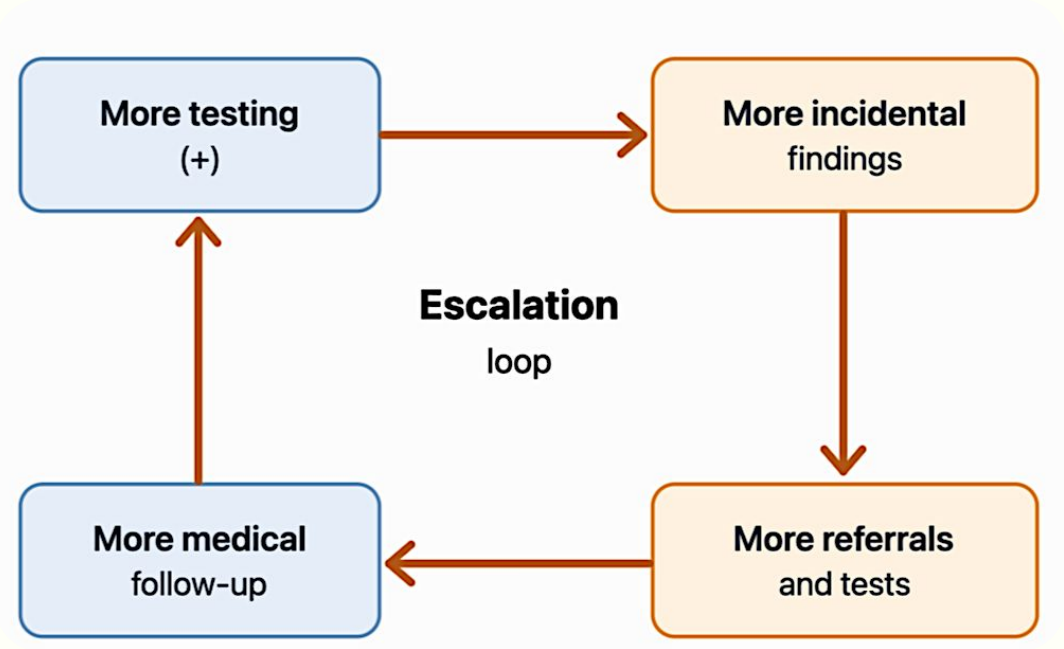


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Was it good advice?

CONTEXT

Low value care and its cascade



Engineering safer care

CONTEXT

ENGINEERING

Before testing

Is the result likely to change patient management?

After an incidental finding

- Risk stratification
- Evidence-based follow-up thresholds
- Evaluate patient outcomes
- Downstream harm and total costs

How alerts become *background noise*

- Humans are error prone
- We need advice
- Why do we ignore it?

→ Because it was potentially harmful

→ Because of its form?

Why advice is ignored

CONTEXT

Types of medical alerts

By sensory modality

- Visual
- Auditory
- Multisensory

By interaction design

- Interruptive (hard stop or soft stop)
- Non-interruptive

Why advice is ignored

CONTEXT

Types of medical alerts

By sensory modality

- Visual
- Auditory
- Multisensory


HR
120
50
75
%SpO2 89
PVC 0
PAP 17/ 14(14)
ABP 118/ 48(68)
HR
120
50
78
%SpO2 93
PVC 0
RESP 14
ST-II 0.4
NBP 113/ 69(79)
HR
120
50
82
%SpO2 93
PVC 0
RESP 19
ST-II 0.2
NBP 113/ 69(79)
Alarms Suspended
HR
120
50
75
%SpO2 ?
PVC 1
RESP 21
ST-II 0.0
NBP 95/ 60(68)

Why advice is ignored

CONTEXT

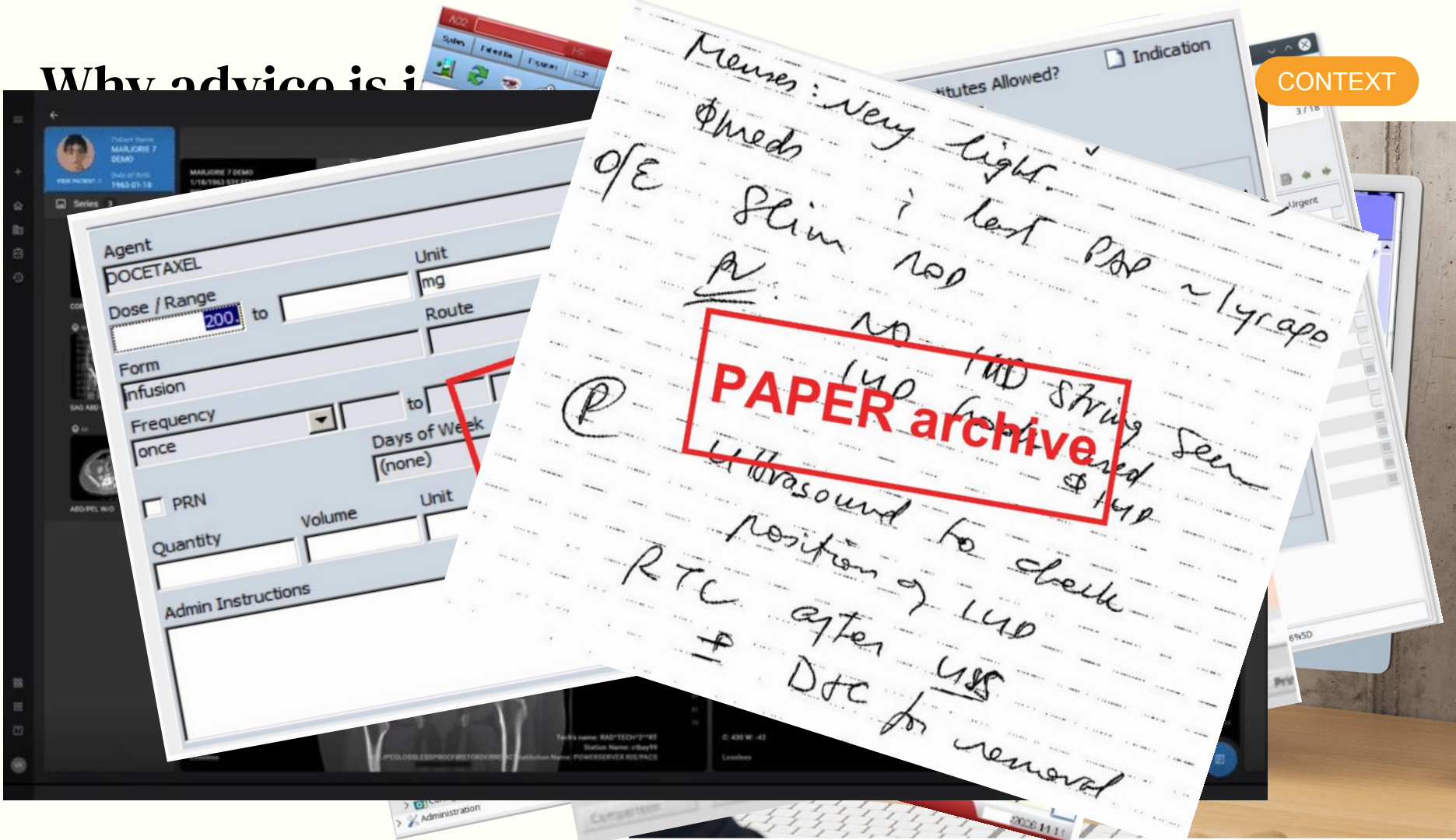
Types of medical alerts

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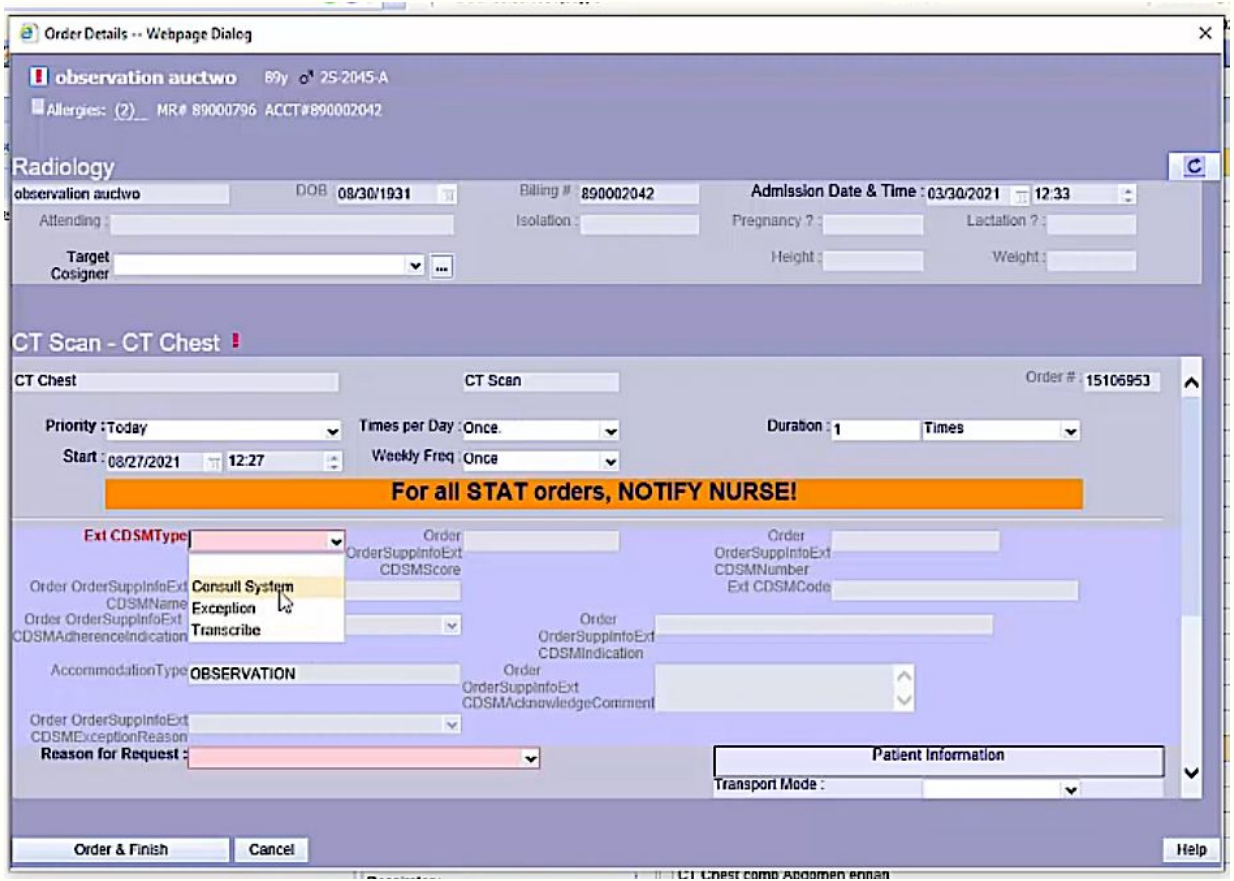
Why advice is i

CONTEXT



Why advice is ignored

CONTEXT

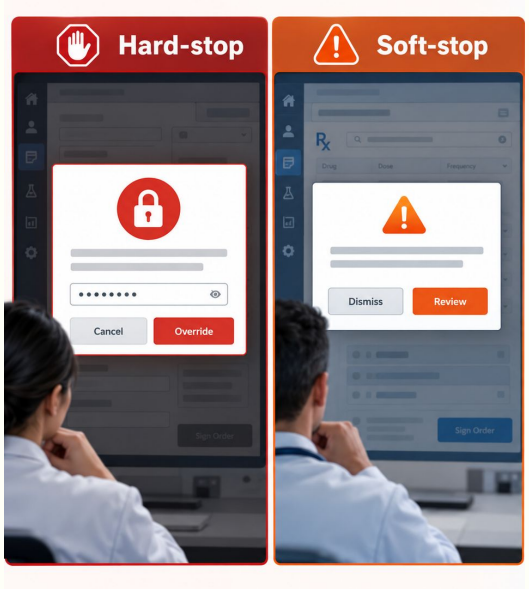


LiGHT

Why advice is ignored

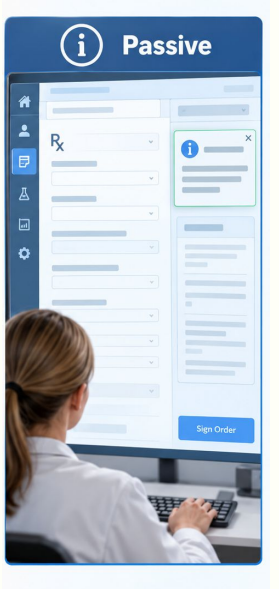
CONTEXT

Types of medical alerts



By interaction design

- Interruptive (hard stop or soft stop)
- Non-interruptive



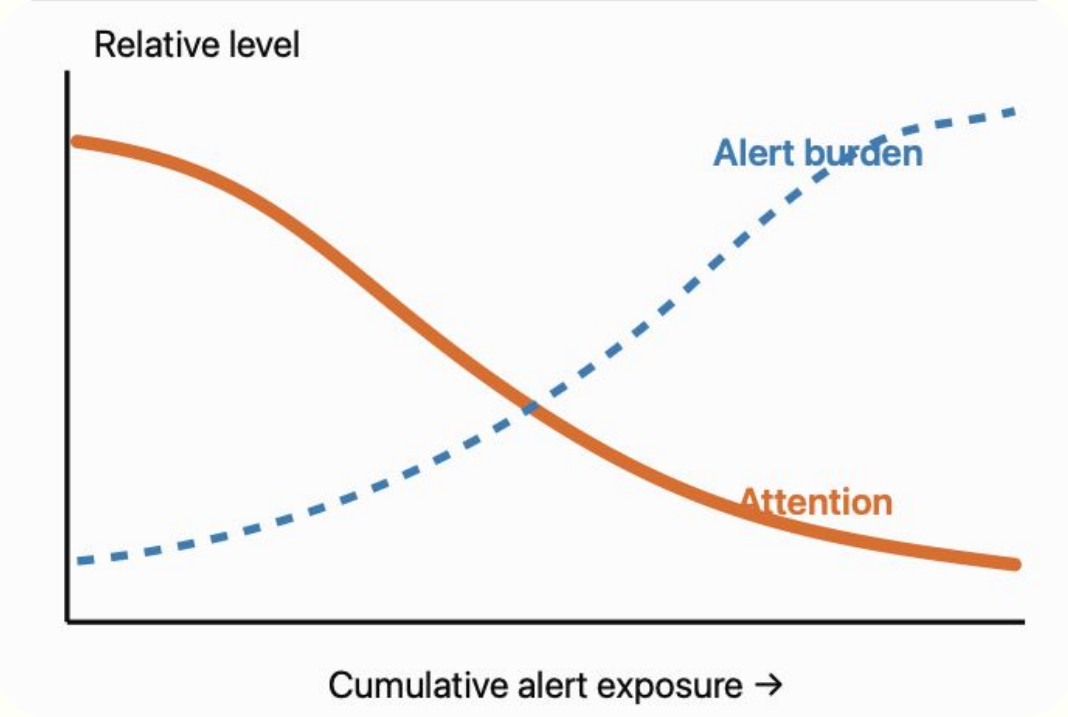
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Why advice is ignored

CONTEXT

Alert fatigue

- Cognitive overload
- Desensitization



Why advice is ignored

CONTEXT

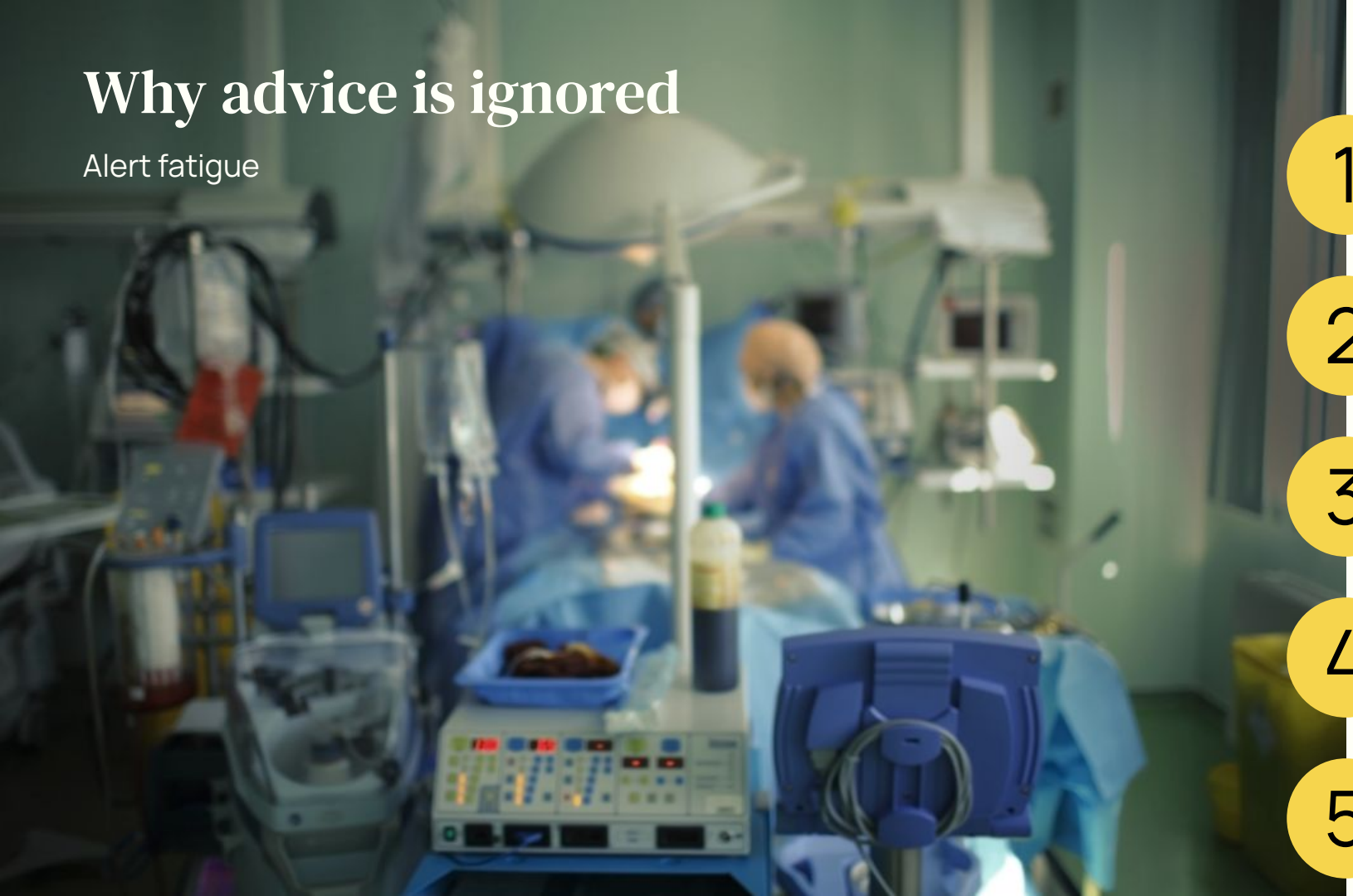
Alert fatigue

Top 3 alerts MDs ignore

- Drug-drug interactions (override rates of 49-96%)
- Preventive care alerts (i.e. vaccines, smoking cessation)
- ICU monitor alarms

Why advice is ignored

Alert fatigue



CONTEXT

1

2

3

4

5

Why advice is ignored

CONTEXT

Other reasons

- Unawareness, unfamiliar or lack of training
- Active disagreement
- Low expected benefit
- Inertia
- External barriers (local guidelines, patient preference)

Co-designing AI for medical advice

Humans are error prone and need advice

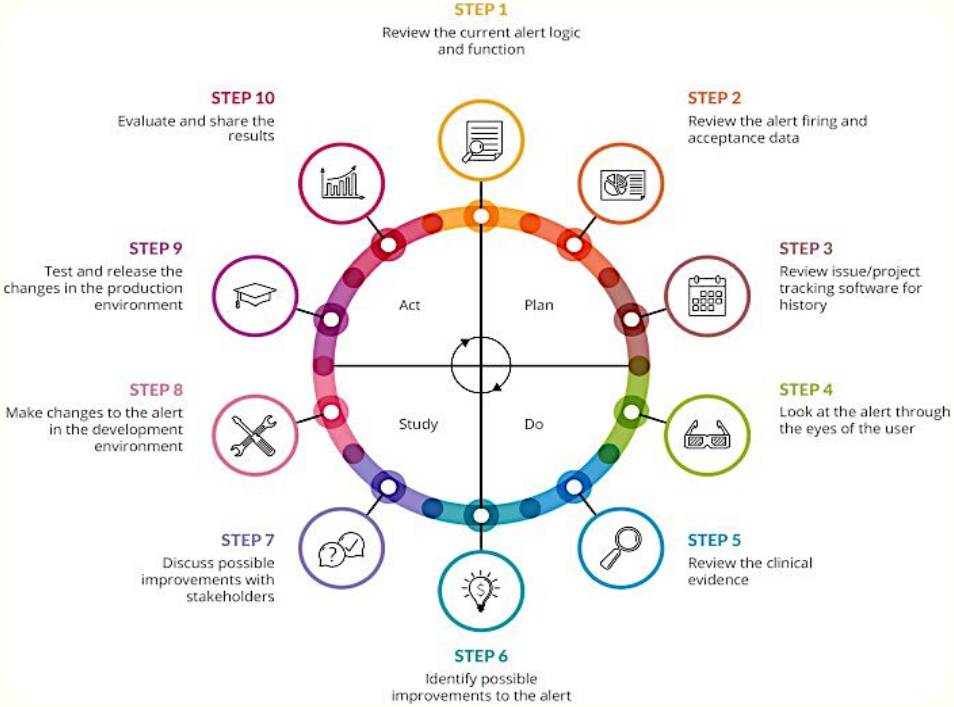
Not every piece of advice deserves a workflow interruption

How can you engineer advice better?

Co-designing medical AI

CONTEXT

Engineer – MD cooperation

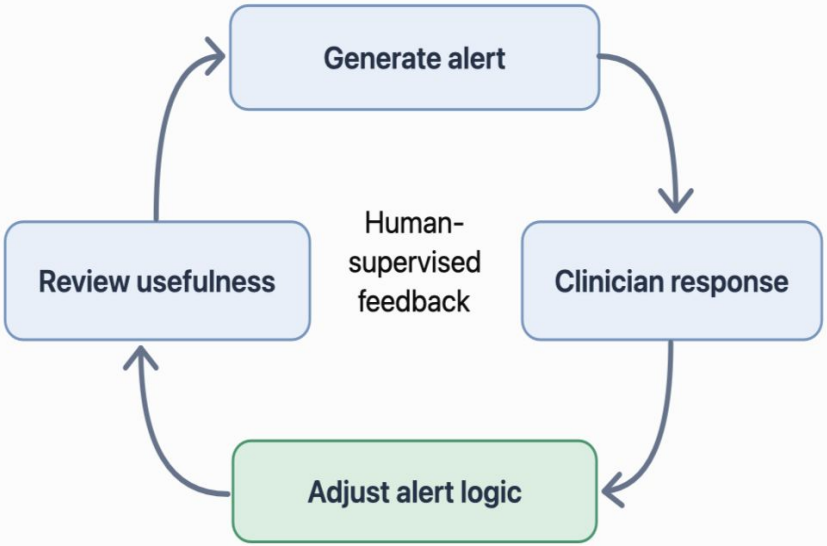


McCoy et al, J Am Med Inform Assoc. 2022

Co-designing medical AI

CONTEXT

Engineer – MD cooperation



Real example:
Vanderbilt's Clickbusters program eliminated approximately 70,000 unnecessary alert firings per week (15%)

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McCoy et al, J Am Med Inform Assoc. 2022

Co-designing medical AI

CONTEXT

Beyond Clickbusters: AI-Driven Alert Optimization

- **From manual to automated curation:**
  XAI/LLMs now scale what Clickbusters did by hand generating alert-optimization suggestions and mining free-text override comments (JAMIA 2022, JAMIA 2024)
- **ML + XAI (Vanderbilt):**
  LightGBM model (2.99M alert firings, AUC 0.919) generated 96 "helpful" suggestions → additional 278,807 firings eliminated (9.3%) beyond manual review (JAMIA 2022)
- **Bonus finding:**
  XAI surfaced workflow/education gaps missed by manual review (JAMIA 2022)
- **LLM-as-judge for order sets:**
  Aligning AI to physician preferences raised agreement from $\kappa = 0.06 \rightarrow 0.41$;
  cut suggestion volume 29% while keeping 92% of useful suggestions (JMS 2024)

Co-designing medical AI

CONTEXT

Engineer – MD cooperation



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Kumar, JAMIA 2020

Summary

More medicine ≠ better medicine
Incidentalomas, low-value care and downstream harm

More alerts ≠ better decisions
cognitive overload and alert fatigue

Context matters
Clinical relevance, right timing, right form

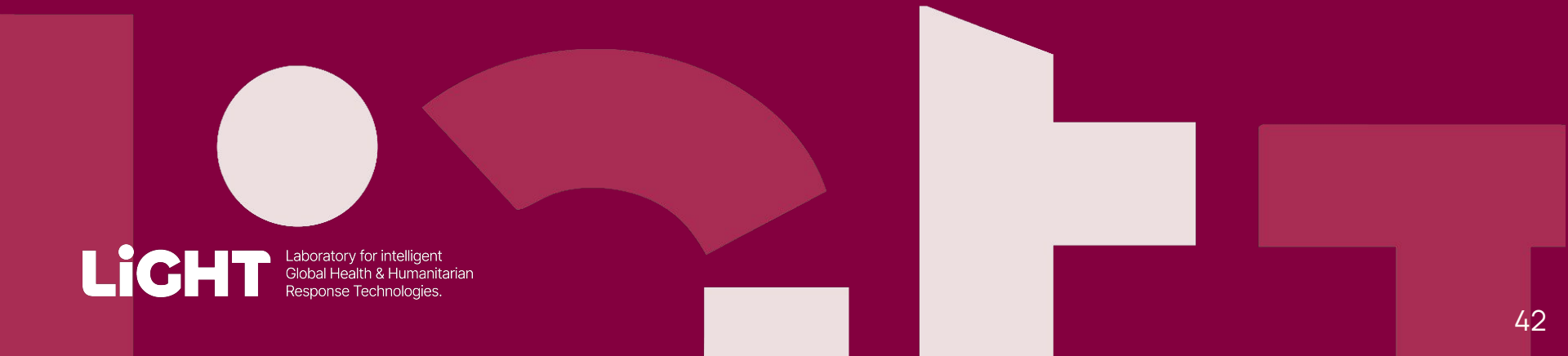
Engineer with clinicians, not just for them
Human-in-the-loop AI design

Questions?

EPFL

Integration Without *Interruption*

Lecturer: *Dr. Lars Klein*



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Attention Is All You Need

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Human attention is
a valuable resource.



Aviation-grade attention management

12. Minimizing the Effects of False and Nuisance Alerts. As much as possible, the alerting functions or system should be designed to avoid false alerts and nuisance alerts, while providing reliable alerts to the flightcrew when needed. The effects of nuisance and false alerts distract the flightcrew, increase their potential for errors, and increase their workload. Section 25.1322(d) requires that an alert function be designed to minimize the effects of false and nuisance alerts. Specifically, a flightcrew alerting system must be designed to:

a. Prevent the presentation of an alert that is inappropriate or unnecessary.

d. Alert Inhibits.

(1) Alert inhibit functions must be designed to prevent the presentation of an alert that is inappropriate or unnecessary for a particular phase of operation (§ 25.1322(d)(1)). Alert inhibits can also be used to manage the prioritization of multiple alert conditions. Inhibiting an alert is not the same as clearing or suppressing an alert that is already displayed.

The Five Rights of CDS

The **right** information

To the **right** person

In the **right** intervention format

Through the **right** channel

At the **right** time in the workflow

Framework source



ahrq.gov

Principles of Mixed-Initiative User Interfaces

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ACM CHI (1999)

ENGINEERING

Eric Horvitz

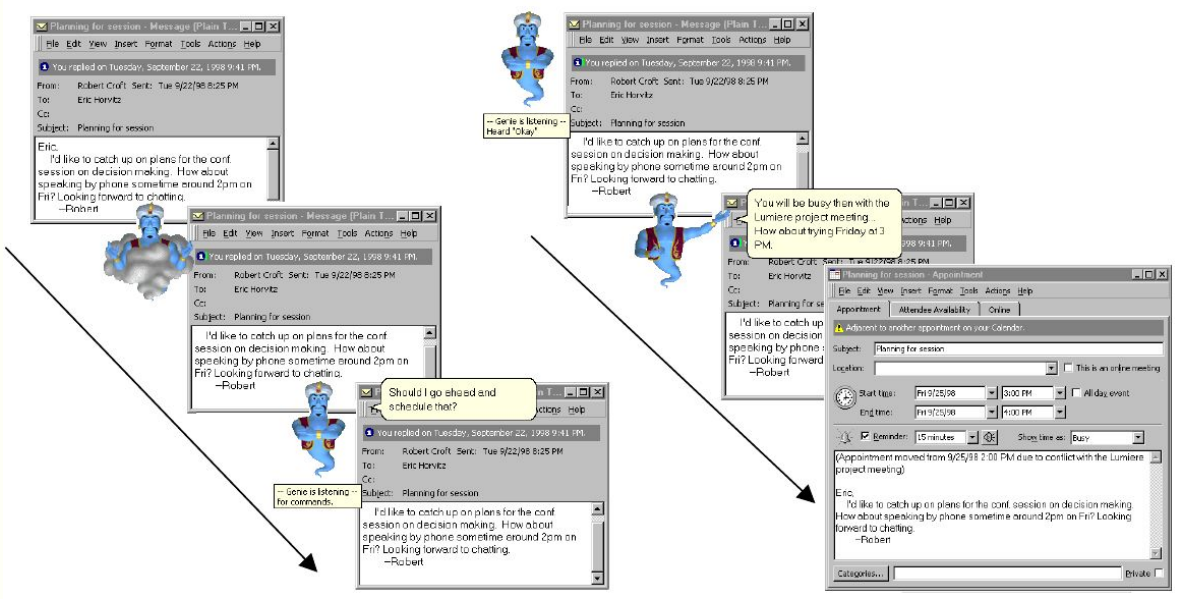
Chief Scientific Officer
Microsoft

MD , PhD

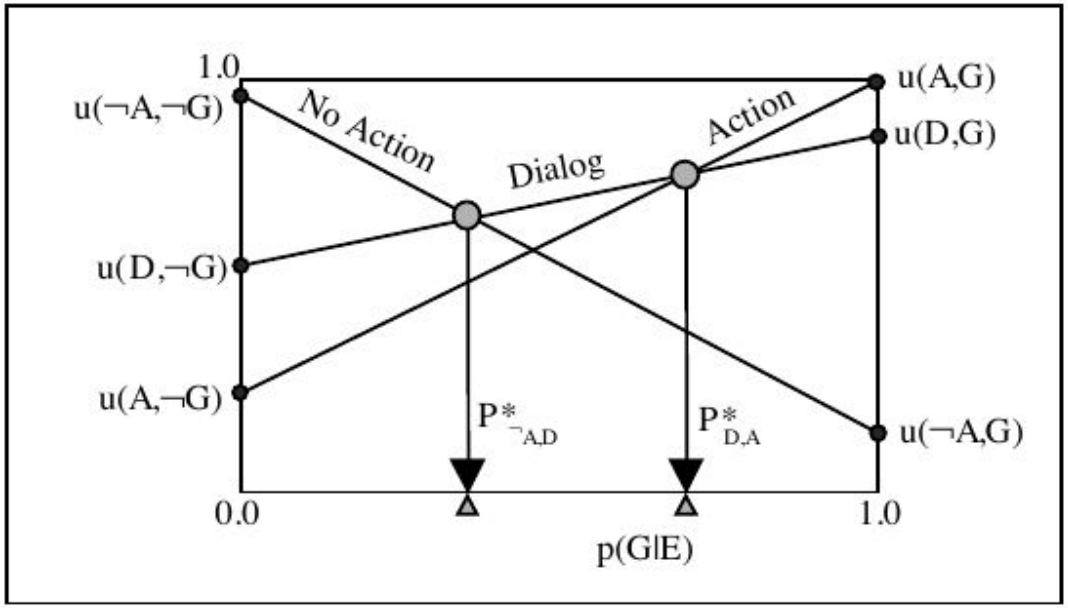


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LookOut (Horvitz, 1999)



Stay quiet, ask, or act?



Journal of the American Medical Informatics Association Volume 10 Number 6 Nov / Dec 2003

Synthesis of Research Paper ■

Ten Commandments for Effective Clinical Decision Support: Making the Practice of Evidence-based Medicine a Reality

DAVID W. BATES, MD, MSc, GILAD J. KUPERMAN, MD, PhD, SAMUEL WANG, MD, PhD, TEJAL GANDHI, MD, MPH, ANNE KITTLER, BA, LYNN VOLK, MHS, CYNTHIA SPURR, RN, MBA, RAMIN KHORASANI, MD, MILENKO TANASIJEVIC, MD, BLACKFORD MIDDLETON, MD, MSc, MPH

Ten commandments for effective CDS

1. Speed is everything
2. Anticipate needs and deliver in real time
3. Fit into the user's workflow
4. Little things can make a big difference
5. Recognize that physicians will strongly resist stopping

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6. Changing direction is easier than stopping
7. Simple interventions work best
8. Ask for additional information only when you really need it
9. Monitor impact, get feedback, and respond
10. Manage and maintain your knowledge-based systems

Ten commandments for effective CDS

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The genie is real.



The genie is real.





The Rise of the AI Co-Pilot:
Lessons for Design from Aviation and Beyond

Abigail Sellen
Microsoft
Cambridge, UK
asellen@microsoft.com

Eric Horvitz
Microsoft
Redmond, WA, USA
horvitz@microsoft.com

Sellen & Horvitz (2024), CACM

Four lessons from aviation

Vigilance

Rare failures are easy to miss.

Monitoring takes effort.

Four lessons from aviation

Vigilance	Taking over
Rare failures are easy to miss. Monitoring takes effort.	Taking control requires knowing what has happened.

Four lessons from aviation

Vigilance	Taking over	De-skilling
Rare failures are easy to miss. Monitoring takes effort.	Taking control requires knowing what has happened.	Skills needed when automation fails can fade through disuse.

Four lessons from aviation

Vigilance	Taking over	De-skilling	Automation bias
Rare failures are easy to miss. Monitoring takes effort.	Taking control requires knowing what has happened.	Skills needed when automation fails can fade through disuse.	Users may accept an answer without questioning it.

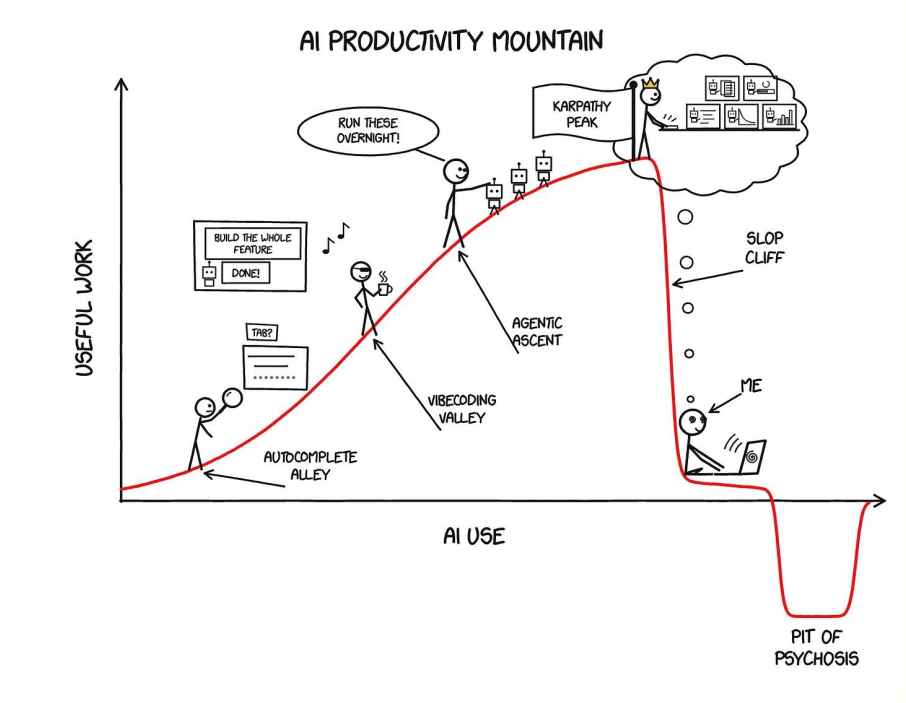
Vibe coding.

Could you take over?

ENGINEERING

Vibe coding.

Could you take over?

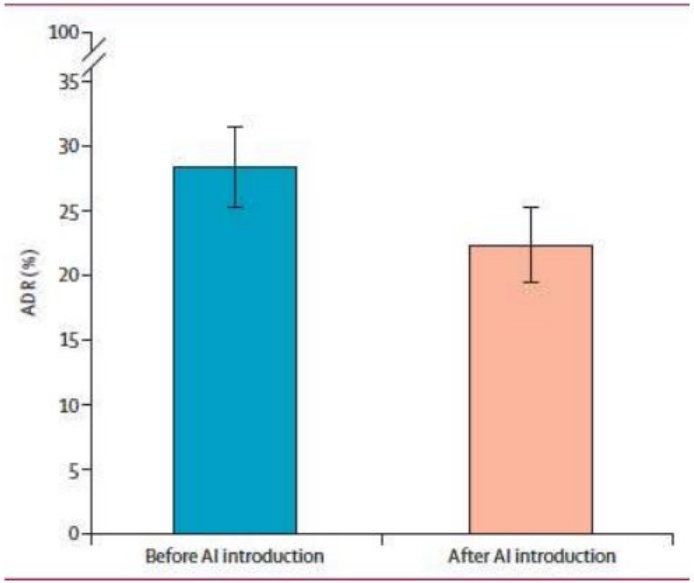


Jiralerspong: Chasing the Karpathy Peak

Deskilling in medicine?

28.4% → 22.4%

Unassisted adenoma detection



Budzyń et al., Lancet Gastroenterology & Hepatology, 2025

nature medicine

Perspective

https://doi.org/10.1038/s41591-026-04438-y

AI-induced never-skilling in medical education

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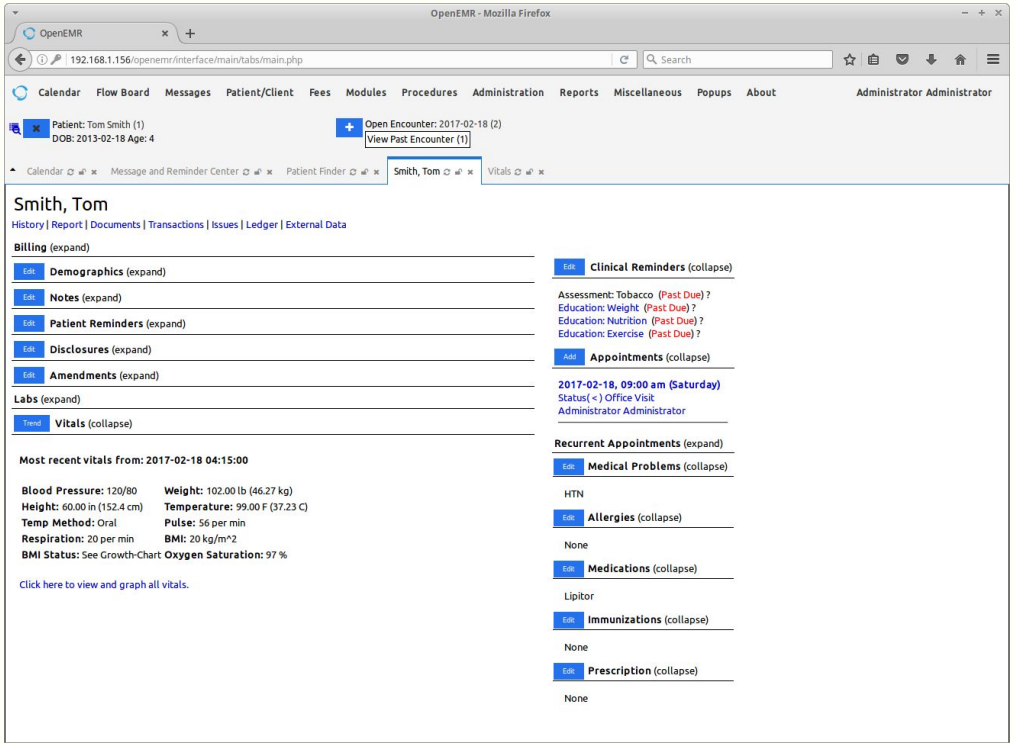
Check for updates

Yuhe Ke[1,2,3], Liyuan Jin[1,4], Jasmine Chiat Ling Ong[1,5], Arun J. Thirunavukarasu[6], Josip Car[7,8], Carol Y. Cheung[9], Yih Chung Tham[4,10,11], Daniel Shu Wei Ting[1,4], Marcus Eng Hock Ong[12,13], Scott Compton[14], Aditee Narayan[15], Pearse A. Keane[16,17], Tien Yin Wong[4,18,19], David W. Bates[20,21,22], Patrick Tan[23,24,25,26] & Nan Liu[1,12,27,28,29]

ENGINEERING

Let’s make it happen.

How do we integrate AI assistants into medical workflows?



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A little vocabulary

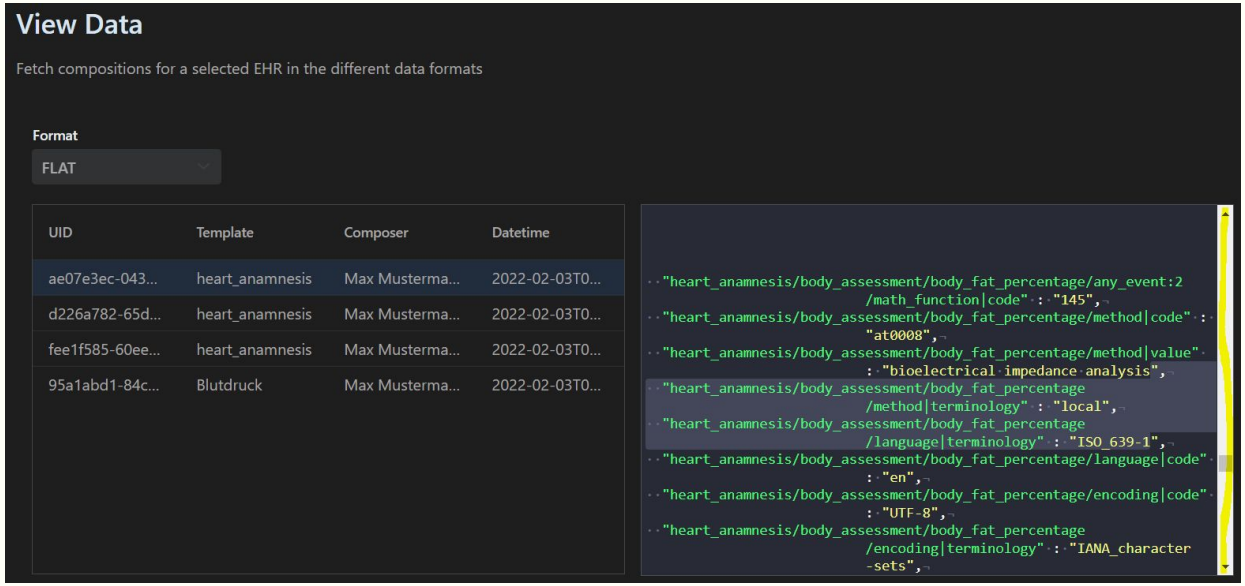
EHR	The clinician's record system
HL7 / HL7 v2	Standards organization / messaging
FHIR	Clinical data and exchange
CDS Hooks	Calls at defined workflow points
openEHR	Clinical record specifications

The published CDS Hooks catalogue

1.2 Hooks

- allergyintolerance-create (1)
- appointment-book (1)
- encounter-discharge (1)
- encounter-start (1)
- medication-refill (1)
- order-dispatch (0)
- order-select (4)
- order-sign (5)
- patient-view (5)
- problem-list-item-create (1)
- medication-prescribe (deprecated)
- order-review (deprecated)

EHRbase



Published sandbox interface

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Global Health & Humanitarian
Response Technologies.

How should advice reach the user?

~~How should advice reach the user?~~

How does the user give us advice?

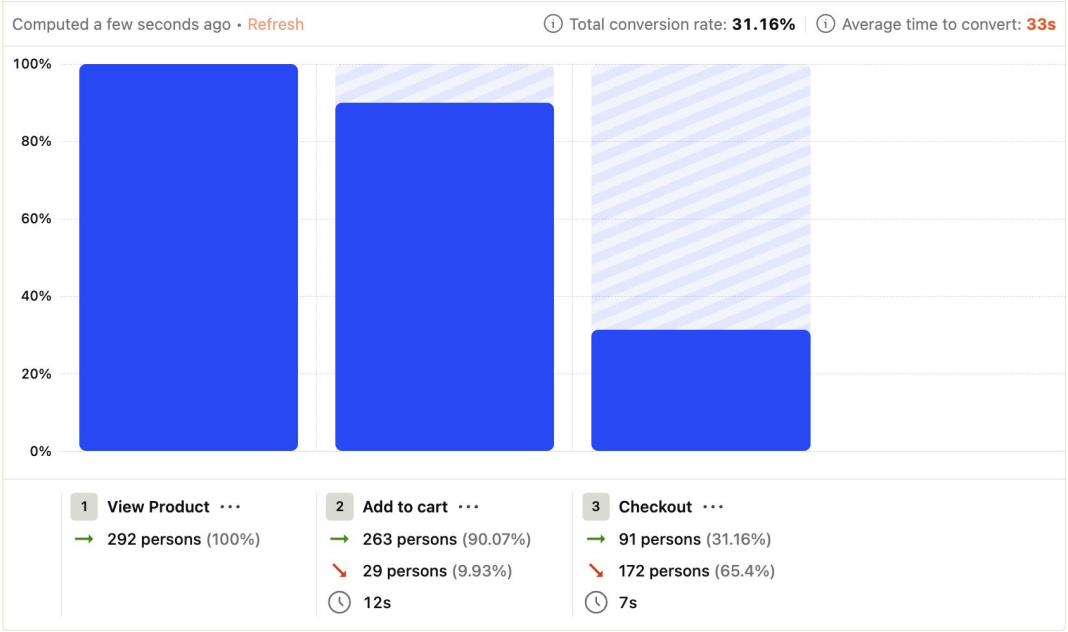
~~How should advice reach the user?~~

How does the user give us advice?

“Continuing to learn by observing.” Horvitz (1999), principle 12



PostHog


Computed a few seconds ago • Refresh
Total conversion rate: 31.16%
Average time to convert: 33s
100%
80%
60%
40%
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1 View Product
292 persons (100%)
2 Add to cart
263 persons (90.07%)
29 persons (9.93%)
12s
3 Checkout
91 persons (31.16%)
172 persons (65.4%)
7s

ENGINEERING

Questions?

Building intelligent Global Health and Humanitarian Response Technologies

Thank You

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Low-value care and the cascade problem

CONTEXT

How big is the problem?



$75.7–101.2B
Estimated annual US spending on low-value care



24–41% Medicare patients receiving ≥1 measured low-value service in a historical study

- More healthcare ≠ better healthcare
- Waste of money, time and clinical capacity
- Potential harm without meaningful benefit

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Bouck, JAMA internal medicine 2020